

Improvement in Food Resource

- Food: Any Organic substance that contains nutrients. i.e; Proteins, Carbohydrate, Fats, Vitamins, Minerals.

- Consumed by living organisms in order to derive energy and support all the vital metabolic activities.

- Food Resource: (Crop)

- A crop is a cultivated population of some type of plants, grown and managed over a larger land area.

1) Food Variety Improvement:

- Focuses on selection of plants with desirable qualities.
 - ↳ Higher Productivity, Increased production per acre.

→ Improved quality = Pulses → Protein
= Oil seeds → oil quality.

→ Increased shelf-life = Fruits and vegetables.

⇒ Biotic and Abiotic resistance:-

- Crop production can go down due to certain biotic factors like - Insects, diseases, nematodes.

→ ~~Abiotic~~ Abiotic factors like - drought, salinity, water lodging, heat, cold, frost.

⇒ Change in maturity duration:-

- ~~Shorter~~ Shorter duration of crops from sowing to harvesting can help farmers to grow multiple rounds of crops.

→ Wider adaptability. Developing various adaptations

pertaining to environment i.e; climate conditions, weather.

⇒ Desirable agronomic trait/character:

- Forage → plant → Tall.
- Food crop → dwarf (short) → because the nutrients utilized for height, will be instead used for production of grains/fruits.

2) Crop Production Management.

- It refers to controlling, organizing and optimizing various practices to ensure maximum yield.
- It is how a farmer manages, nutrients water supply and cropping techniques to help healthy and

abundant growth of crops.

→ Nutrient Management

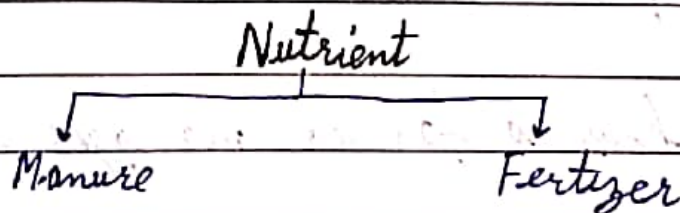
- There are various types of nutrients required by plant.

⇒ Among these are the nutrients required in large quantities, these are known as macronutrients.

e.g. Carbon, hydrogen, oxygen, ~~the~~ (CHO), (NPK) → Nitrogen, Phosphorus, Potassium.

⇒ While as the nutrients required in lesser amount are known as micro-nutrients.

e.g. Copper, zinc, Nickel



⇒ Manure: It is a natural organic substance, produced by decomposition.

- Substances like plants and animals wastes.

⇒ Types of Manure:

1) Compost: It is prepared by decomposing agricultural waste, household wastes (Vegetable peel) and live stock wastes.

2) Vermi-Compost: Compost prepared by using earth worms.

⇒ Green Manure: The Fast-growing plants are grown and then ploughed and mulched with soil. e.g. Sun-hemp.

⇒ Benefits: It provides

- Nutrients
- Increase water holding capacity.
- Improves texture of soil and aeration
- Reduces reliance on synthetic chemicals which causes soil pollution.

⇒ Fertilizer:

• These are commercially produced synthetic substances or chemical compound.

e.g. NPK fertilizer.

Urea fertilizer.

→ Easy to absorb.

→ Target essential nutrient requirement.

⇒ However excessive use of fertilizers cause harm to soil microbes.

→ Change PH of soil.

→ Washing out of these nutrients into water bodies cause eutrophication.

- Nutrient enrichment of water bodies.
- Growth/excessive growth of algae.
- Depletion of water oxygen i.e; dissolved

oxygen.

⇒ P.H.:- PH means Potential of hydrogen ion.

• A scale used to measure, how acidic, basic a substance is.

⇒ Decomposition:- Break down of substance, by microbes/micro-organism.

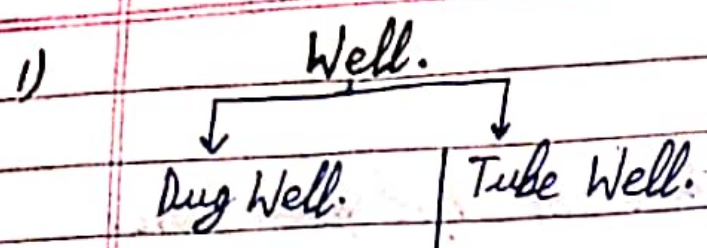
⇒ Irrigation/Water supply:-

• Indian agriculture depends on monsoon for water supply.

• But the crops require water growing period.

• Different method for water supply.

- 1) Well. (2) Canals. (3) River/riverlift system.
- 4) Tanks.



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| <ul style="list-style-type: none">• This well draws water from bearing strata/foundation strata. | <ul style="list-style-type: none">• It draws water from deeper strata/layers.• Water is uplifted with the help of the pump. |
|--|--|

2) Cannal

- Extensive and elaborate water supply system.
- Cannal divides into branches.
- And branches into field.

3) River Lift System

- When Cannal has inadequate supply of water then the water is directly drawn from rivers.

4) Tank: Small storages, that collect run-off

water from catchment areas.

→ Crop Pattern

- Different ways of growing crops, in order to get maximum benefit and yield.
- It is of following types:

a) Mixed Cropping

- Grow two or more crops on same piece of land without following any specific pattern.
- Benefit This method reduces risk or acts like Insurance against complete crop failure.

e.g. - Wheat + Gram
Wheat + Mustard

b) Inter-Cropping - Growing two or more crops

together on same land but following a specific pattern of rows.

- Benefit This method ensures maximum utilization of nutrients.
- It prevents ~~spread~~^{spread} of disease to all the crop plants.

e.g. Maize + Soybean.

Cow pea (^{lobia}~~bajra~~) + finger millet (^{bajra}~~lobia~~).

→ Crop rotation -

- Growing different crops on same land, but in pre-planned succession year after year.

→ Benefit - Growing different crops yearly ensures fertility of that particular ~~year~~ land area.

⇒ Crop Production Management - ...

- It refers to all agricultural practices and complete set of decision by a farmer, during cultivation and storage after harvesting.

⇒ Crops can be damaged by:

- 1) Weeds
- 2) Insects
- 3) Diseases.

1) Weed - It is an unwanted plant.

- Impact - It competes with:- nutrients, space, sunlight.
→ over all growth of crops is reduced.
e.g Parthenium (Gajar Grass).

2) Insects - They damage crops by three methods.

- 1) By cutting root, stem and leaves.
- 2) Draw sap from different parts of plant.
- 3) Boring into stem and ~~leaves~~ fruit.

3) Plant Diseases. A plant gets diseased by bacteria, Viruse, Fungi.

- These diseases spread through, soil, water, air.

⇒ Preventive and control measures:

a) Chemical method~

Spraying of pesticides

Herbicide. Insecticide. Fungicide.

Pest~ Any Animal/Insect that damages crop.
cide~ To Kill.

(b) Mechanical or cultural method:

- Removal of Weeds by hand.
- Proper seed bed making.
- Timely Sowing.
- Crop rotation
- Inter cropping.

c) Preventive measure.

- Use of disease resistant varieties.
- Summer ploughing.

(4) $\Rightarrow$ Storage of Grains

- High storage ~~loss~~ losses can drastically reduce the ~~production~~ produce quality and marketability.

$\Rightarrow$ It can occur by following factors.

- 1) Biotic factors: Bacteria, fungi, mites, rodents.
- 2) Abiotic factors: Inappropriate moisture content and temperature during storage.

$\Rightarrow$ Effects of storage degradation:-

- Quality degradation.

- Weight loss.
- Discolouration
- Poor germination
- Low market value.

⇒ Prevention and control measures:

- Strict cleaning: Thoroughly cleaning of ^{before} grains $\uparrow$ storage.
- Drying: First sundrying, follow by shade.
- Fumigation: Using safe chemical to kill pests.
- It is a pest control method in which sealed space is filled with ~~gas~~ gaseous chemicals, in order to eliminate pests.

⇒ Animal husbandary (Like Stock).

- It is the ~~of~~ scientific management of animal

live stock (feeding, breeding, disease control).

- ~~It~~ It includes = ~~Crop~~ Csu.

= Goat.

= sheep

= Poultry

= Fishes

= Honey bees.

⇒ Need- Due to increasing population.

- Rising living standards.

- Increased demand for meat, milk and eggs.

⇒ Cattle Farming:- Two Reasons/Purpose:-

1) Milk → Milch animals

2) Agricultural purposes → Draught Animals.

*⇒ Lactation:- The period of milk production after birth of young one (calf)

★ → Cow: *Bos Indicus*. (Scientific Name)

★ → Buffalo: *Bos Bubalis*. (Scientific Name)

⇒ Indigenous Varieties of Cow

- Red Sindhi
- Sahiwal.

⇒ These varieties have disease resistance.

⇒ Exotic Varieties: High milk production.

- Jersey
- Brown Swiss

⇒ Cross breeding

- Breeding done between exotic and indigenous

varieties, in order to get both desirable qualities

i.e. ~~the~~ disease and resistance and high milk yield.